Quantifying Mechanisms of HIV Immunotherapy Using Models of Post-Treatment Viral Rebound

Madeleine S. Gastonguay[1*], Mélanie Prague[2], Alison Hill[1,3]

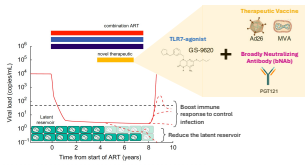
[1]Institute for Computational Medicine, Johns Hopkins University, [3]

Introduction

Motivating question: How does immunotherapy alter HIV rebound kinetics?

HIV can be treated but not cured with cART

The current standard of care for HIV is life-long combination antiretroviral therapy (cART). If treatment is stopped, viral rebound typically occurs within weeks due to an ineffective immune response against re-activation of latently infected cells. Research into HIV cure strategies is therefore focused on immunotherapies that might delay or prevent viral rebound by reducing the latent reservoir and/or enhancing the antiviral immune response.

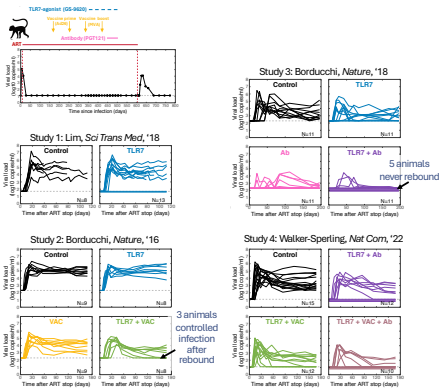


Preclinical studies have demonstrated induction of SIV or SHIV remission by immunotherapy

Three immunotherapies have been evaluated in non-human primate (NHP) models of HIV[1-4]:

- TLR7-agonist, which activates the antiviral innate immune response
- Therapeutic vaccine, which is given during infection
- Broadly neutralizing antibody (bNAb) PGT121, which neutralizes diverse HIV strains

Rhesus macaques infected with SIV (studies 1&2) or SHIV (studies 3&4) received cART alone or with single/combination immunotherapy, and viral rebound was monitored after treatment washout. Many animals exhibited altered viral rebound kinetics, and a subset achieved remission:

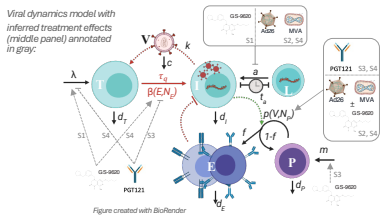


Goals:

- Determine if each therapy works by reducing the latent reservoir and/or boosting the antiviral immune response
- Identify synergistic drug effects
- Predict the efficacy of single and combination immunotherapy in preventing or delaying viral rebound in people with HIV

Mathematical Model and Parameter Inference

We analyzed rebound data following immunotherapy from the above NHP studies using the basic viral dynamics model extended to include latent cell reactivation and a generalized antigen-driven immune response.



For each NHP study, we estimated the distribution of individual parameter values governing latent reactivation (t_a), immune effector activation/expansion (m, p, N_p), target cell availability (λ), and viral infectivity (β) using mixed-effects modeling. This approach allows us to **determine which of these processes each therapy modulates and test for drug synergies** by identifying statistically significant differences in mean parameter values between treatment groups (*treatment effects*).

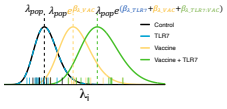
$$\log(\lambda_i) = \log(\lambda_{pop}) +$$

Effect of drug 1: $\beta_{\lambda_TLR7}[TLR7 = 1] +$

Effect of drug 2: $\beta_{\lambda_VAC}[VAC = 1] +$

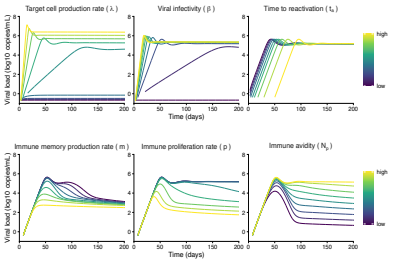
Synergistic effect: $\beta_{\lambda_TLR7:VAC}[TLR7:VAC = 1] +$

Random Effect: $\eta_{\lambda,i} \quad \eta_{\lambda,i} \sim N(0, \Omega_\lambda)$



We identified significant treatment effects through BICc-guided covariate model building approaches modified to reduce overfitting and identify uncertainty in inferred treatment effects (more details in right column). In addition, we evaluated pre-defined hypotheses of drug mechanisms and selected the model that best describes the data.

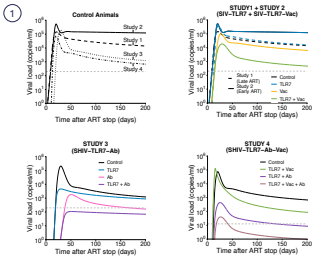
Qualitative Trends in Rebound Data are Captured by the Model



Univariate sensitivity analysis illustrates that target cell availability and viral infectivity have similar influences on early exponential growth and peak viremia. Immune proliferation and avidity both influence set-point viremia.

Mechanistic Insights to HIV Immunotherapies

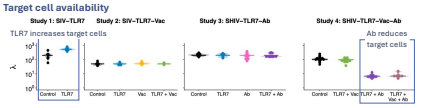
Our analysis identified significant differences in several biological processes ②-⑦ that can explain difference in rebound trajectories following immunotherapy ①.



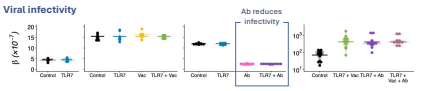
Individual immunotherapeutic agents can act simultaneously on the latent reservoir and the immune response:

- Therapeutic vaccination reduces latent reactivation ④ and enhances effector activation ⑦
- bNAb reduces viral infectivity in one cohort ③ and target cell availability in another ②, enhances effector activation ⑦, and has no significant effect on the latent reservoir ④
- TLR7-agonist effects are context-dependent:
  - With late ART initiation, it reduces reactivation ④ and increases target cell availability ②
  - With early ART, it increases immune memory in one cohort ⑤ and synergizes with vaccination to boost effector activation in another ⑦

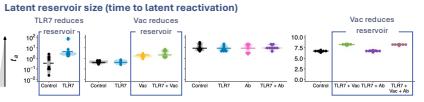
② **Target cell availability**



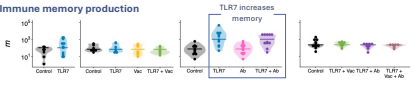
③ **Viral infectivity**



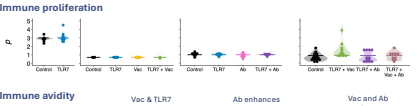
④ **Latent reservoir size (time to latent reactivation)**



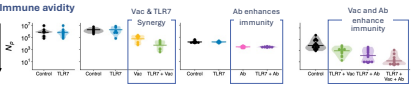
⑤ **Immune memory production**



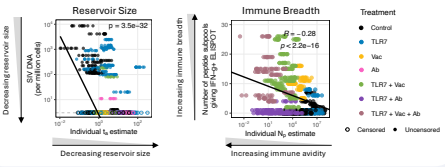
⑥ **Immune proliferation**



⑦ **Immune avidity**

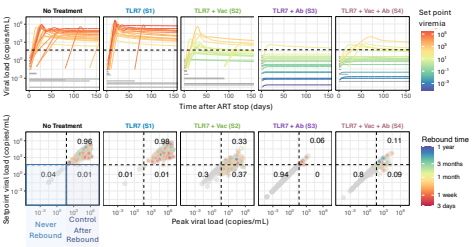


We found independent support of these effects in strong correlations between individual parameter estimates of reservoir size (t_a) and immune avidity (N_p) with SIV/SHIV DNA in PBMCs and the breadth of the immune response to *gag*, *pol*, and *env* peptides at treatment interruption.



Predicting Efficacy of Immunotherapy in Human Trials

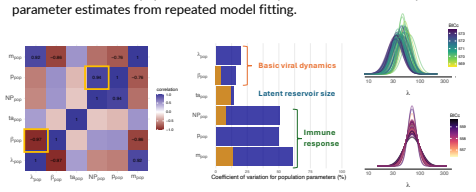
To predict the ability of single or combination therapy to establish HIV remission in humans, we conducted virtual clinical trials. We simulated human trials by fitting the model to HIV rebound data in the absence of therapy, sampling parameters from the posterior, and applying NHP-derived treatment effects.



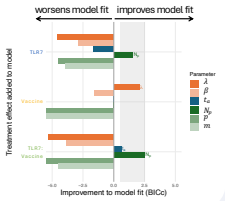
Combination therapy including bNAb induces durable viral control in the majority of virtual patients

Assessing Robustness of Parameter Inference

We evaluated practical identifiability of model parameters via repeated model fitting with different initial guesses[5-7] and found strong correlations between two viral dynamics parameters and two immune response parameters. Fixing their mean values by profile likelihood improved the consistency of all parameter estimates from repeated model fitting.



To identify significant treatment effects, we used a covariate model building approach that can be prone to overfitting and may exclude models that provide similarly good model fits from consideration. We implemented this method with safeguards against overfitting and a comparison of the best-fit model to alternative models defined from intermediate steps.

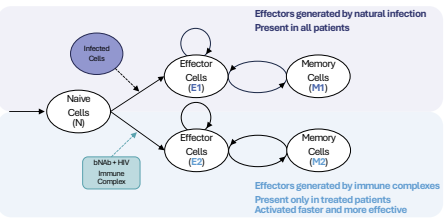


Key Findings

- Both therapeutic vaccination and bNAb enhance the immune response to control infection below detectable levels
- The effect of the vaccine on the reservoir is enhanced by the TLR7 agonist
- Both TLR7-agonst with late ART initiation and therapeutic vaccination can reduce the latent reservoir
- Innate immune activation by the TLR7 agonist may act against its therapeutic potential by increasing target cell availability, but can also lead to increased immune memory
- BNAb slows the rate of early growth upon viral rebound by reducing viral infectivity or target cell availability
- We did not detect synergy between bNAb and the TLR7 agonist in reducing the latent reservoir, contrary to our hypothesis
- Combination therapy, especially regimens with bNAb, may induce HIV remission in patients
- Patients may rebound before establishing control, indicating that clinical trials evaluating HIV cures may require longer ART interruption windows or altered restart criteria to capture remission.

Future Directions

How do bNAbs enhance the antiviral immune response?



References: [1] Lim et al., *Sci. Trans. Med.*, 2018 [2] Borducchi et al., *Nature*, 2016 [3] Borducchi et al., *Nature*, 2018 [4] Walker-Sperling et al., *Nat. Com.*, 2022 [5] Duchesne et al. *BMC Bioinform.*, 2021 [6] Lavielle et al., *JPKPD*, 2016 [7] Cassidy et al., *Bull. Math. Biol.*, 2026

*email: mgaston1@jh.edu